Symbiosis Institute of Technology, Pune.

Department of Artificial Intelligence and Machine Learning

Automation Project Report

Batch 2023-27

Continuous Assessment – III (CA-3)
Academic Year: 2026–2027

Project Title: Automated Expense Claim Validation and Notification using n8n

Student Name(s): Varun Mate ; Vedant Shitole

PRN: 23070126142 ; 23070126143

Division: B3

Course Name: Automation Project

Faculty Name: Dr. Priya Jadhav

Submission Date: 23 October 2026

Project Summary

The project introduces a system for automated expense claim validation and alerts created using n8n Cloud, Google Sheets and Gmail. The existing manual expense management system requires administrators to review claims, verify amounts, detect repetitive claims, update records and inform applicants about the outcome. The repetitive nature of the task means delays in responses and inconsistencies in records.

The system in question collects claim ID, employee name, email address alongside the amount, category and an optional description through an online form. As soon as the form is submitted, the claim goes through an event-driven workflow. The first step is a decision-making process, whereby the amount is checked for being greater than 0. In case the amount is invalid, an email with an explanation of the rejection is sent. If the amount is valid, the claim gets checked against the records in Google Sheets through the claim ID. If there is a matching ID, a duplicate claim rejection is triggered. In case of a unique claim, it is added to the Claims sheet with status, submission time and execution ID. Then, Gmail provides an acknowledgement of the storage.

An Audit Log sheet registers the result, rationale, claim information, timestamp and execution identifier for every finished business process. The three processes were tested by means of a valid claim, a negative amount and duplicate claim identifier. The combined audit test resulted in one successful claim record as well as three audit entries, together with the thus defined notifications.

The project provides evidence of visual workflow engineering, conditional routing, interfacing with services and due business exception handling. The software performs the intake as well as the first stage of validation; as for all other functions such as approval by the manager, verification of the receipt and reimbursement – they are not included in the software's functions.

1 Problem Statement

An employee expense claim usually passes through several stages of administration before it can be reimbursed. In the current manual approach, employees review each claim, check relevant values, verify past claims, log the accepted information in a spreadsheet and issue a reply. This burdening process leads to excessive workloads and to inefficient handling of wrong and double claims.

The business issue is to automate this submission process yet keep an accurate record of accepted and rejected claims. The solution ought to check positive amounts claimed, avoid repeating Claims ID of the already processed ones and indicate to the submitter what the result is.

2 Objectives

- Collect expense requests through a structured form and trigger processing on submission.
- Validate that the amount is greater than zero and provide a correction message for invalid values.
- Search existing claims by Claim ID and route duplicates away from record creation.
- Record valid unique claims and send an acknowledgement using the configured Gmail account.
- Maintain an audit trail and demonstrate successful and exception paths using multiple scenarios.

3 Scope of the Project

The procedure consists of budgeting, validating the amount, finding the Claim ID, storing the accepted claim, sending the email notifications, and documenting the result. The form has to have all of the fields filled up except for the Description. The amounts are stated in INR. Google Sheets is used as the permanent repository of records. The communication is organized through Gmail.

The procedure requires that the Google account is in use, and the Internet connection is available. The Claim ID is considered to have been successfully performed as the unique id. The difference in the IDs of the same expense is not revealed. The Category is given name. The software does not verify the identity of the employee as the sender of the claim or the receipts, company policy, approval, and payment.

4 System Requirements

Category	Requirement
Hardware	A laptop or desktop capable of running a modern browser; no GPU required.
Software	A modern browser such as Chrome or Edge and internet connectivity.
Development platform	n8n Cloud workspace with form, If, Google Sheets and Gmail nodes.
Cloud services	Google Sheets for records and audit entries; Gmail for messages.
Accounts and access	n8n account and Google credentials with access to the selected spreadsheet and sending mailbox.
Local installation	No local workflow runtime or separate backend installation is required.

5 System Architecture

The n8n Cloud is an orchestration layer that receives events from forms, evaluates conditions, invokes Google services and transmits outputs. The spreadsheet Expense Claim Records has two sheets; Claims stores accepted claims and Audit Log stores done business results. Three nodes of Gmail give details of the outcome of all the routes.

There are 11 nodes in the workflow: one form trigger, two If nodes, two operational Sheets nodes, three Gmail nodes and three audit append nodes. The history of execution provides another view of the operations giving timings, conditions of branching, input data and mistakes.

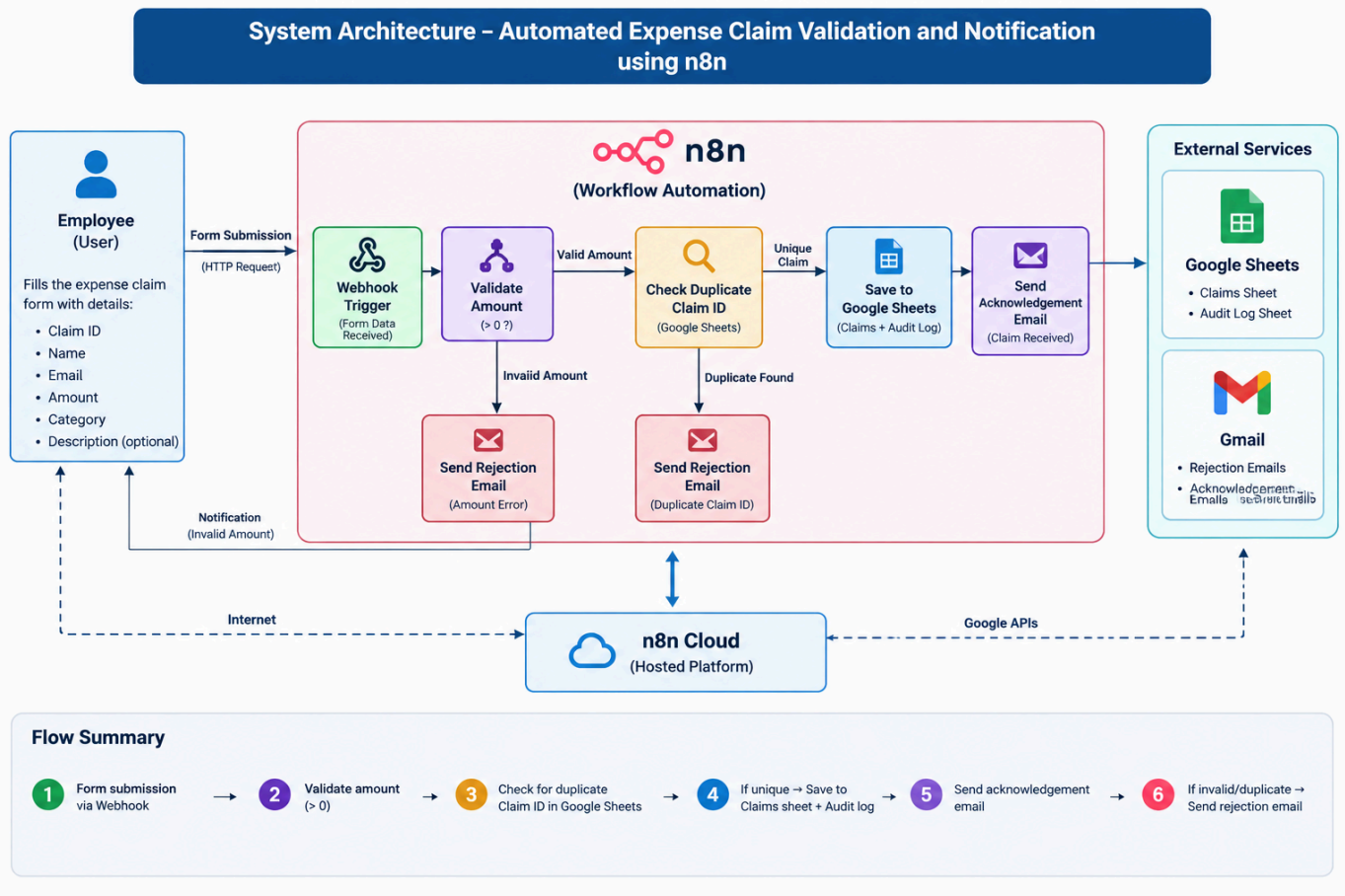


Figure 1 Final workflow architecture with all three audit paths

Audit entries follow the notification nodes. Consequently, a failed email prevents the downstream audit append on that path. The design provides traceability for completed business paths; it is not a guaranteed audit of every failed service call.

6 Automation Workflow

Node	Action and decision
On form submission	Collect Claim ID, Employee Name, Email, Amount, Category and Description.
Validate Amount	Compare numeric Amount with 0. True continues to lookup; False sends rejection.
Check Duplicate Claim	Get Row(s) from Claims, filtered by Claim ID. Always Output Data is enabled.
Duplicate Found?	Check whether the returned Claim ID is nonempty. True indicates a duplicate.
Save New Claim	Append the original form values, Submitted status, timestamp and execution ID.
Send Claim Acknowledgement	Send a receipt confirmation after the claim append succeeds.
Reject Invalid Amount	Send the reason and request a corrected positive amount.
Reject Duplicate Claim	Explain that the Claim ID already exists and no new claim is created.
Three audit nodes	Append Submitted or Rejected outcome and reason to Audit Log.

Decision logic and data preservation

```
Validate Amount: {{ Number($json.Amount) }} > 0
Lookup filter: {{ $json["Claim ID"] }}
Duplicate Found: {{ $json["Claim ID"] ?? "" }} is not empty
```

An empty Sheets lookup normally has no matching row to pass forward. Always Output Data emits an empty item so the duplicate decision still executes. Subsequent nodes read the original form node explicitly to preserve submitted values. This avoids relying on the empty lookup item or the Gmail response payload.

Retry strategy

The implementation uses business exception branches and execution-history inspection. Automatic retries and a separate service-error recovery branch are not configured. Recovery should resume from the failed stage after checking saved records; blindly rerunning an append can duplicate a write.

7 Technology Stack

The platform n8n Cloud keeps the visual workflow and forms. The JavaScript expressions will fetch fields, convert amounts and create timestamps for us. The connectors from Google Sheets and Gmail will leverage the Google APIs via OAuth authentication. No use of a custom Python server, SQL database, or separate frontend is needed.

8 Implementation

Form and validation

The title of the form is Employee Expense Claim. In this form, Claim ID, Employee Name and Category are text fields; Email is an email input; Amount is a number input; and the Description uses a textarea. In the workflow amount condition, there is a clear rejection route for both negative values and values of zero. Once the form is submitted, it is recognized by the system; therefore, the submission page acknowledges that the form has been received and that it has not been processed yet.

Records and audit schema

Tab	Columns
Claims	Claim ID; Employee Name; Email; Amount; Category; Description; Status; Submitted At; Execution ID
Audit Log	Execution ID; Claim ID; Employee Name; Amount; Outcome; Reason; Timestamp

Claims uses Append Row after the duplicate decision returns False. Audit nodes also use Append Row so separate submissions remain separate events. Append or Update Row is avoided for the audit trail because a matching record could be overwritten.

Expressions and notification settings

```
Claim ID: {{ $('On form submission').first().json["Claim ID"] }}
Email: {{ $('On form submission').first().json.Email }}
Submitted At: {{ $('On form submission').first().json.submittedAt }}
Execution ID: {{ $execution.id }}    Audit timestamp: {{ $now.toISO() }}
```

All three Gmail nodes use the verified credential named Vedant Gmail. Its connected college mailbox is the sender, while the form Email field determines the recipient. The acknowledgement includes the Claim ID, amount, category and Submitted status. Rejections give the specific reason. Credential names identify connections; renaming a credential does not change its mailbox.

Security and deployment

Google OAuth authentication is done from n8n and is not included inside the expressions and reports. Without authentication in this version, email validation is often confused for authentication. Exports need to protect any credentials or private data (payload).

Deployment process: Verify three pathways: publish the workflow, copy the Production URL for the form, and make an unused claim without running the process yourself. Confirm the Claims, audit, and acknowledgement records. Production activation is required in addition to the manual testing performed.

9 Test Cases

The following results describe observed executions and the user-confirmed combined audit test. Test identifiers refer to scenarios; additional edge checks are separated from completed tests.

Test ID	Scenario	Expected Output	Actual Output	Status
TC01	Valid claim EXP003; 750 INR	Append one claim and send acknowledgement.	Spreadsheet updated and email received.	Passed
TC02	EXP002; amount -100	Use invalid branch and send rejection.	False branch executed; rejection email received.	Passed
TC03	Repeat existing EXP001; 500 INR	Use duplicate branch; send notification.	Existing row returned; duplicate email received. Sender corrected and retested.	Passed
TC04	Fresh EXP005; 600 INR with audit	One accepted claim, email and Submitted audit entry.	Included in combined test; accepted row and notification confirmed.	Passed
TC05	EXP006; -100 INR with audit	Rejected audit entry and invalid notification.	Included in combined test; rejection outcome logged.	Passed
TC06	Repeat EXP005 with audit	Rejected audit entry; accepted count unchanged.	Combined test confirmed one claim row and three audit rows across TC04–06.	Passed

Additional boundary and service tests

Zero value, required fields missing, invalid email format, optional empty Description, Google API error and execution of published Production URL are beneficial acceptance tests. These tests were absent from the confirmed results above. Recovery from service failure needs to be tested independently in case implemented.

Verification method

For every route in business, it is important to see what active nodes are there in actual execution history, to compare email reason with its input and to see what are the destination tabs. When there’s a remark, compare the number of matching Claims rows before and after submission. You can expect an audit entry for the attempt regardless of whether any claim was recorded.

10 Exception Handling

Condition	System response	Recovery
Amount is zero or negative	Send invalid-amount rejection; append Rejected outcome after successful notification; bypass Claims append.	Correct the amount and resubmit.
Claim ID already exists	Send duplicate rejection; append Rejected outcome; bypass new claim creation.	Check the ID or refer to the existing claim.
No matching claim row	Always Output Data supplies an empty item; duplicate condition is False.	Continue to append the new claim.
Google authentication or API failure	The execution can fail and retain diagnostic details in n8n history. No custom recovery branch is claimed.	Restore access or service, inspect prior writes and retry the failed stage carefully.

Business rejects are known processes rather than crashes. Technical failures are operational errors. If the Claims append is successful and Gmail fails, the claim is saved but the follow-up audit append is not processed. The claim would then be entered into the form and processed as duplicative instead of sending the original acknowledgement again.

11 Performance Evaluation

Functional assessment highlights the focus on proper routing, validity of records and notifications delivered. The combined auditing exercise resulted in just a single accepted claim record and 3 result records for valid submissions, negative amounts and duplication submissions. These records reflect the expected performance of the examined series rather than prove reliability statistically.

Measure	Observation or interpretation
Observed invalid-run duration	One saved invalid-amount execution completed in 732 ms.
Accepted-record integrity	One accepted row across the three-scenario audit test.
Audit coverage	Three audit entries for three completed business paths.
Notification verification	Acknowledgement and both rejection types were received.
Scalability boundary	No load test performed; Google API limits and concurrent lookup–append operations constrain scale.

The 732 ms value is a single execution-history observation, not a mean response time or inbox delivery latency. No percentage reduction in manual effort, guaranteed throughput or long-run availability is asserted. A broader benchmark should record multiple complete executions under a defined workload.

12 Results

The workflow interlinks form input with the stages of validation, record storage, and employee communication. The successful full path test updated Google Sheets and sent an acknowledgment notification. The negative amount test produced a False output from the Validate Amount step. The duplicate test retrieved the existing row of the spreadsheet and sent a duplicate notification.

Every business path is complemented with the audit logging. The combined test that included a valid claim, an invalid amount, and an already used Claim ID produced only one Claims record and three Audit Log records. The execution identifier connects every audit entry with its n8n run.

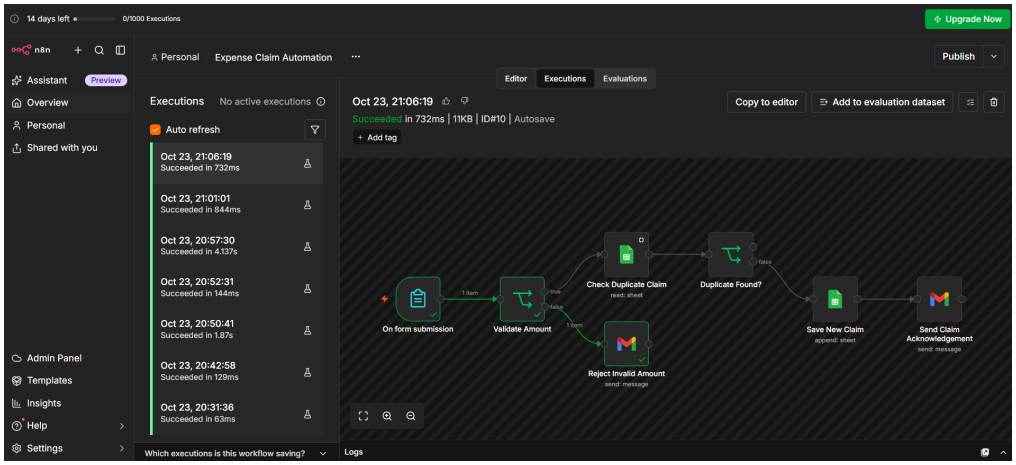


Figure 2 Execution history shows the invalid amount branch and successful rejection delivery

13 Innovation and Future Enhancements

One of the roles of engineering is that they merged many features designed for validating conditions, searching for duplicates, notifying employees, and taking record of the output into one simple-to-read flow. The ability to reference data source nodes allows keeping form data despite empty searches and responses from APIs. The use of execution IDs allows connecting business data with operational process history.

The update plan involves introducing manager approvals, uploading receipts and OCR, implementing category limitations, protecting employee access, keeping track of notification states and limiting retries. Having a proper database designed in a way to ensure that every Claim ID is unique would help implement safe duplicate prevention during concurrent processing operations.

14 Conclusion

The presented system showcases automation of expense input and pre-validation. The working claims will be processed creating a record of accepted and rejected claims based on the business rules and approved validity criteria. The testing confirmed the business workflow and audit information generated by it. The displayed concept allows one to learn how to design the workflow, write expressions, create service accounts, map data and direct exceptions. After the approval, reimbursement and production hardening can be added to the information shown.

16 Annexure

A Duplicate notification evidence

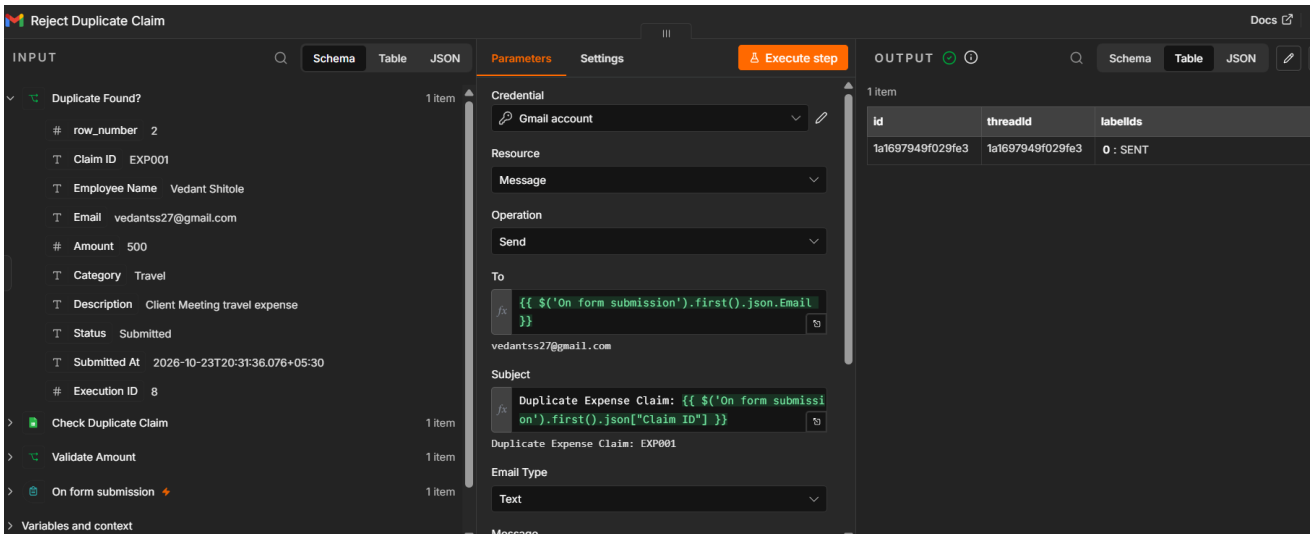


Figure 3 Duplicate claim notification returned SENT using the corrected Gmail connection

B Audit record definitions

Outcome	Reason
Submitted	Claim recorded and acknowledgement sent
Rejected	Amount must be greater than zero
Rejected	Duplicate Claim ID

Each audit row also records Execution ID, Claim ID, Employee Name, Amount and Timestamp. The Timestamp uses the workflow runtime clock in ISO format, including timezone information. Audit rows record submission attempts; Claims rows represent accepted submissions.

C Operational limitations

The audit sheet is editable and is not a tamper-proof ledger. Logging after email covers completed paths but may miss notification failures. A public form can receive unwanted submissions. Duplicate matching does not validate receipt authenticity and does not enforce a transaction across lookup and append.

16 Annexure continued

D Demonstration sequence

Time	Demonstration content
0:00–0:35	State the expense-intake problem and existing manual process.
0:35–1:10	Show the workflow architecture and Google Sheets/Gmail connections.
1:10–2:10	Submit a fresh valid claim; verify Claims, acknowledgement and audit record.
2:10–2:55	Submit a negative amount; show rejection and audit entry.
2:55–3:40	Repeat an existing Claim ID; show duplicate rejection and unchanged claim count.
3:40–4:25	Inspect execution history, execution IDs and audit reasons.
4:25–5:00	Discuss scope, limitations and future enhancements.

E Workflow and repository references

Workflow name: Expense Claim Automation

Workflow source: export the final n8n JSON as Workflow/Expense_Claim_Automation.json.

Repository link:https://github.com/vedants27/expense-claim-automation

Production form URL: https://vedantca3.app.n8n.cloud/form/dca8c3bb-5c05-4af6-aecc-a19a9fd1bb57

F Technical reference links

n8n documentation: https://docs.n8n.io/

Google Sheets API documentation: https://developers.google.com/workspace/sheets/api

Gmail API documentation: https://developers.google.com/workspace/gmail/api